INCH-POUND
MIL-DTL-223D
26 March 2022
SUPERSEDING
MIL-DTL-00223C(AR)
05 February 1993
MIL-P-223B
26 February 1962

DETAIL SPECIFICATION

POWDER, BLACK

This specification is approved for use by all Departments and Agencies of the Department of Defense within the distribution limitations noted at the bottom of the page.

1. SCOPE

1.1 Scope. This specification covers black powder and is intended for use in ammunition.

1.2 Classification. The black powder furnished under this specification is of the designated classes analogous to the corresponding nominal granulations as specified in Table III.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army Armaments Center, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000, or emailed to usarmy.pica.devcom-ac.list.ardec-stdzn-branch@army.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

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COMMERCIAL ITEM DESCRIPTION

A-A-59138	-	Charcoal

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-155	-	Graphite, dry
MIL-DTL-156	-	Potassium nitrate
MIL-DTL-14929	-	Sulfur, commercial grade

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-286	-	Propellants, Solids; Sampling, Examination and Testing
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product
MIL-STD-3010	-	Test Procedures for Packaging Materials

(Copies of these documents are available online at https://quicksearch.dla.mil/).

U.S. ARMY DEVCOM AC DRAWINGS

7548077	-	Container, Metal, Packing for Black Powder, Assembly
7548078	-	Container, Metal, Packing for Black Powder, Details
7548321	-	Drum Steel Packing for Black powder, Assembly Details, Packing and Marking
7548322	-	Drum Steel Packing for Black Powder, Details
8863394	-	Primer, Percussion, M92E1, Loading and Assembling
12982865	-	Minimum Marking Instructions for Ammunition and Explosives Containers

(Copies of these drawings are available via email usarmy.pica.ccdc-ac.list.drawing-request-help-desk@army.mil.)

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BUREAU OF NAVAL WEAPONS

LD299212	-	Container, Powder, MK 3 Mod 0

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Contracting Officer).

2.2 Non-Government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D257	-	Plastic: Method of Testing
ASTM E11	-	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves
ASTM E70	-	Standard Test Method for pH of Aqueous Solutions with the Glass Electrode

(Copies of these documents are available from https://www.astm.org/.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subject to conformance inspection in accordance with 4.3.

3.2 Material. The constituent material used in the manufacture of black powder shall comply with the requirements specified in Table I.

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TABLE I. Material.

CONSTITUENT MATERIAL	CONFORMING TO:
Graphite	MIL-DTL-155, Grade C (see 6.7)
Potassium nitrate	MIL-DTL-156, Class 1
Charcoal	A-A-59138, Class 1 (see 6.7)
Sulfur	MIL-DTL-14929

3.3 Composition. The composition of the black powder shall comply with the requirements specified in Table II.

TABLE II. Composition.

Constituent	Class 1 through 7, and Class 9 (%)	Class 8 only (%)
Potassium nitrate	74 ± 1.0	74 ± 2.0
Sulfur	10.4 ± 1.0	10.4 ± 1.5
Charcoal	15.6 ± 1.0	15.6 ± 1.5

3.4 Moisture content.

3.4.1 Applicable to class 2, and classes 4 through 9. The moisture content of the black powder shall be no greater than 0.70 percent.

3.4.2 Applicable to class 1 only. The moisture content of the black powder shall be no greater than 0.60 percent.

3.4.3 Applicable to class 3 only. The moisture content of the black powder shall be no greater than 0.30 percent (see 6.14).

3.5 Ash content. The ash content of the black powder shall be no greater than 0.80 percent.

3.6 Specific gravity.

3.6.1 Specific gravity determined by mercury pycnometer. The specific gravity of black powder shall not exceed the limits of 1.72 minimum (min.) and 1.80 maximum (max.) for glazed powder, 1.65 min. and 1.81 max. for unglazed class 7 powder and 1.69 min. and 1.76 max. for unglazed class 8 and 9 powder.

3.6.2 Specific gravity determined helium gas pycnometer. The specific gravity of black powder shall not exceed the limits of 1.890 min. and 2.012 max. for classes 1 to 6 glazed powder and 1.955 min. and 1.997 max. for unglazed class 7 powder.

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3.7 Apparent density. The apparent density of the black powder shall be based on its use in ammunition as follows:

3.7.1 General bulk loading. For general bulk loading of components in which the weight-volume ratio is not critical, the apparent density (applicable to classes 1 and 2 only) shall be 1.03 grams per milliliter (g/mL) min.

3.7.2 Close weight-volume tolerance. The apparent density of classes 1 to 6 may be specified as follows: For specific applications involving close weight-volume tolerances, the apparent density of classes 1 and 2 shall be 1.10 g/mL min., and the apparent densities of classes 3 to 6 inclusive shall be 1.03 g/mL min. and 1.10 g/mL max. (see 6.2).

3.8 Granulation. The granulation for the applicable class of black powder shall be as specified in Table III.

TABLE III. Granulation.

Class	Retained on U.S. Standard Sieve		Passing through U.S. Standard Sieve	
	sieve #	% max.	sieve #	% max.
1	4	3	8	5
2	6	3	12	5
3	8	3	16	5
4	16	3	30	5
5	16	3	40	5
6	20	3	50	5
7	40	3	100	7
8	100	5	270	50
9 [1]				

Note 1: 128 ± 2 grains/pound, 0.6 inch diameter sphero-hexagonal shaped.

3.9 Gritty or fibrous particles. The powder shall be free of all gritty or fibrous particles.

3.10 Glaze finish (not applicable to classes 7, 8, or 9). Glazed black powder shall have a consistent glazed appearance as a result of application of graphite in accordance with MIL-DTL-155.

3.11 Functioning (see 6.7 and 6.12). The glazed black powder, when loaded in a M92E1 primer, shall function 100 percent.

3.12 pH. The pH of black powder shall be between 6.0 min. and 8.0 max.

3.13 Workmanship. The black powder lot shall be homogeneous through the lot, and of uniform texture / color, free from any foreign materials and contaminants. If any sample fails to meet any test requirement, the lot represented by the sample shall be rejected.

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4. VERIFICATION

TABLE IV. Requirements/verification cross reference matrix.

Method of verification 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test						Classes of verification A – First article B – Conformance		
Section 3 Requirement	Description	Verification Methods				Verification Class		Section 4 Verification
		1	2	3	4	A	B	
3.1.1	First article		X	X	X	X		4.2
3.1.2	Conformance			X	X		X	4.3
3.2	Material			X	X	X	X	4.4.3
3.3	Composition				X	X	X	4.4.3
3.4	Moisture				X	X	X	4.4.2
3.5	Ash				X	X	X	4.4.5
3.6	Specific gravity				X	X	X	4.4.6
3.7	Apparent density				X	X	X	4.4.7
3.8	Granulation				X	X	X	4.4.8
3.9	Gritty or fibrous particles				X	X	X	4.4.4
3.10	Glaze finish				X	X	X	4.4.1
3.11	Functioning				X	X		4.4.9
3.12	pH				X	X	X	4.4.10
3.13	Workmanship			X		X	X	4.4.11
3.13	Packaging			X		X	X	4.3.3.2 / 4.3.3.3

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see paragraph 4.2)
b. Conformance inspection (see paragraph 4.3)

4.2 First article inspection. When specified (see 6.2), a sample shall be subjected to first article verification in accordance with Table V. Sampling procedures shall be conducted in accordance with 4.3.2. Samples shall be produced by the contractor using the same process, procedures, material supply sources, production equipment, and production line as will be used in regular production.

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TABLE V. First article acceptance tests.

Category	Examination or Test	Requirement Paragraph	Inspection Method
Critical	None defined		
Major			
101	Potassium nitrate determination	3.3	4.4.3
102	Sulfur determination	3.3	4.4.3
103	Charcoal determination	3.3	4.4.3
104	Moisture	3.4	4.4.2
105	Ash	3.5	4.4.5
106	Specific gravity	3.6	4.4.6
107	Apparent density	3.7	4.4.7
108	Granulation	3.8	4.4.8
109	Gritty or fibrous particles	3.9	4.4.4
110	Glaze finish	3.10	4.4.1
111	Functioning	3.11	4.4.9
112	pH determination	3.12	4.4.10
Minor			
201	Workmanship	3.13	4.4.11

4.2.1 Rejection. If any sample fails to comply with any of the applicable requirements the first article quantity shall be rejected.

4.3 Conformance inspection.

4.3.1 Inspection lot formation. A lot shall consist of one or more batches of black powder, produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall contain black powder of one class only. Each batch shall consist of that quantity that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous.

Each lot shall contain:

a) Graphite from one manufacturer with one lot number.
b) Potassium nitrate from one manufacturer with one lot number.
c) Charcoal from one manufacturer with one lot number.
d) Sulfur from one manufacturer with one lot number.

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4.3.2 Black powder sampling. Sampling for inspection shall be performed at either the supplier/distributor location where the black powder is packed, or at the manufacturer's receiving facility where the material is unpacked for use in the next higher sub-assembly or formulation. There shall be no reduced or tightened inspection level. If any sample fails to comply with any applicable requirements the batch shall be rejected.

4.3.3 Classification of characteristics. Black powder shall be subjected to the inspections according to 4.3.3.1.

4.3.3.1 Black powder.

Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Potassium nitrate determination	4.3.2	3.3	4.4.3
102	Sulfur determination	4.3.2	3.3	4.4.3
103	Charcoal determination	4.3.2	3.3	4.4.3
104	Moisture	4.3.2	3.4	4.4.2
105	Ash	4.3.2	3.5	4.4.5
106	Specific gravity	4.3.2	3.6	4.4.6
107	Apparent density	4.3.2	3.7	4.4.7
108	Granulation	4.3.2	3.8	4.4.8
109	Gritty or fibrous particles	4.3.2	3.9	4.4.4
110	Glaze finish	4.3.2	3.10	4.4.1
111	pH	4.3.2	3.12	4.4.10
Minor				
201	Workmanship	VL II	3.13	4.4.11

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4.3.3.2 Container prior to sealing.				
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major	None defined			
Minor				
201	Liner improper	LV II	3.13	Visual

4.3.3.3 Container sealed.				
Category	Examination or Test	Verification Level	Requirement Paragraph	Inspection Method
Critical	None defined			
Major				
101	Weight maximum	LV I	3.13	5.1
102	Improperly sealed	LV I	3.13	Visual
Minor				
201	Marking misleading or unidentifiable	LV II	3.13	Visual

4.3.2.4 Sampling for pH. Take a sample from each completed glazing barrel batch for one pH determination. If the sample fails to comply with the requirement of 3.12, the batch shall be rejected.

4.3.2.5 Sample preparation for analysis. A 1 ounce portion of the primary sample shall be ground in a suitable mortar to pass through a No. 60 U.S. Standard sieve complying with ASTM E11 (Class 8 need not be ground or screened unless, upon examination, the powder exhibits lumps or clusters). All precautions shall be taken to avoid unnecessary exposure of the sample to the air; hence, as soon as a portion is ground, it shall be placed in a bottle and tightly stoppered. If the grinding and sifting does not require more than 3 minutes per portion, there will be no appreciable change in the moisture content due to hygroscopicity. Additional 1 ounce portions shall be ground as needed. The powdered sample shall be mixed well before analysis.

4.4 Methods of inspection.

4.4.1 Glaze finish. The black powder shall be examined visually to ascertain that it has been glazed with graphite and that surfaces of the grains do not have a mottled appearance, which may indicate inadequate glazing or high moisture content.

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4.4.2 Moisture. From the ground portion of the primary sample, a weighed portion of approximately 2 g shall be transferred to a tared weighing dish or covered watch glass and dried for 4 hours at 70 degrees Centigrade (C) to 75 degrees C. The dish shall be covered, cooled in a desiccator and weighed. The loss in weight shall be calculated in terms of percent moisture. (As an alternative method, the portion may be dried for not less than 72 hours at room temperature in a desiccator over concentrated Sulfuric acid). Three such determinations shall be made and the average of the three results reported as the percent moisture in the batch.

4.4.2.1 Alternate method A for moisture. Exactly 10 g of bulk black powder or pressed pellets shall be weighed in an aluminum disposable pan liner, properly positioned, on the Ohaus Moisture Determination Balance. The infrared heat lamp that is built into the machine shall be moved directly over the powder sample, heat adjusted to the desired temperature, and time and wattage controls regulated to a setting based on a predetermined drying curve. The moisture determination shall then be read directly in either grams or percent moisture loss with the accuracy of the final percent moisture or percent moisture loss based on the accuracy of the initial 10 g sample. The pan and dried powder shall be weighed until constant weight is obtained, or until the weight between weighings does not exceed one milligram.

4.4.2.2 Alternate method B for moisture. The moisture shall be determined in accordance with MIL-STD-286, Method 101.4.

4.4.2.3 Alternate method C for moisture. An Arizona Instrument Corporation's Computrac MAX-50 moisture analyzer or its equivalent shall be used for this method. The Computrac MAX-50 moisture analyzer is a microprocessor controlled instrument with programmable heating temperatures from 25 degrees C to 225 degrees C which incorporates a closed loop heat sensing system that will maintain temperature control within plus or minus 1 degrees C. Self-diagnostics assure that a Fail Safe condition will exist, shall a default occur. The analyzer heats the sample within an enclosed test chamber with the use of a nichrome heating element. Through a predictive feature, the analyzer automatically terminates the test at an accurate end point. A built-in electronic force balance automatically weighs the sample as it is being added then displays the final moisture content as either 'percent moisture' or 'percent solid' at test completion. Resolution of the MAX-50 is 0.01% and is designed to accurately detect moisture levels down to 0.10% (10 gm capacity unit) and 0.04% (20 g capacity unit).

The unit (Computrac MAX-50 or its equivalent) shall be turned on and allowed to warm up for about 20 minutes. Set the unit to display '%MOIST'. The test temperature shall be set at 85 degrees C. Open the test chamber lid and place a clean sample pan onto the tray support. With the test chamber lid closed, push the 'START' key. When the 'LOAD' light on the display illuminates, open the test chamber lid, and evenly deposit enough black powder onto the sample pan until the display reads between 70 and 90 percent at which time the analyzer will emit a short beep to alert the operator that enough sample has been added. Close the test chamber lid and the test will automatically proceed. The analyzer will emit three short beeps and display '%MOIST', 'FINAL', and 'CHECK' at the completion of the test. The moisture content of the sample shall be read directly from the '%MOIST' display on the analyzer."

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4.4.3 Composition.

4.4.3.1 Potassium nitrate content.

4.4.3.1.1 Gravimetric method. From the ground portion of the primary sample, a weighed portion of approximately 10 g shall be transferred to a 400 mL beaker and 200 mL of distilled water added. The beaker shall be placed on a steam bath and the contents brought to a boil and held for 15 minutes. The contents of the beaker shall be filtered through a tared filtering crucible and washed with successive portions of 10 to 15 mL of the water. The bath water passing through the crucible shall be tested with an excess of concentrated Sulfuric acid containing a few crystals of diphenylamine, until there is no blue color, (a blue color indicates the presence of nitrate). The crucible shall be dried for 4 hours at 70 degrees C to 75 degrees C or until all moisture is removed, cooled in a desiccator, and weighed. The loss in weight represents moisture and Potassium nitrate. The percent Potassium nitrate shall be calculated on a moisture free basis, utilizing the result obtained in 4.4.2 as the percent moisture. The crucible and contents shall be reserved for the determination of Sulfur (see 4.4.3.2).

4.4.3.1.2 Nitrometer method. The Potassium nitrate shall be determined in accordance with MIL-STD-286, Method 209.3. In case of dispute or doubt, the Nitrometer method shall be used as the standard method.

4.4.3.2 Sulfur content. After the determination of Potassium nitrate, as specified in 4.4.3.1, the crucible shall be placed in an extractor on a water bath, and extracted for 4 hours with Carbon disulfide. After the extraction, the crucible shall be washed once with Alcohol and once with Ether, using suction, dried for 1 hour at 100 degrees ± 3 degrees C, cooled in a desiccator, and weighed. The loss in weight shall be obtained by subtracting this weight from the weight of the crucible and contents, after extraction, obtained in 4.4.3.1 and the result calculated to percent Sulfur on a moisture-free basis. The crucible and contents shall be reserved for the determination of charcoal (see 4.4.3.3).

4.4.3.3 Charcoal content. The residue remaining in the crucible, reserved from 4.4.3.2, shall be considered to be charcoal. Its weight shall be determined by subtracting the tare weight of the filtering crucible from the total weight of the crucible and residue. This weight shall be calculated to percent charcoal on a moisture-free basis.

4.4.4 Gritty or fibrous particle content. The residue remaining in the crucible, reserved from 4.4.3.3, shall be visually examined to observe that it is entirely free from all gritty or fibrous particles.

4.4.5 Ash content. The crucible with its residue (see 4.4.3.3) shall be ignited in a muffle furnace or over a Bunsen burner until all the Carbon is burned off, cooled in a desiccator and weighed. The weight of residue shall be calculated as percent ash. If it is desired to perform this analysis simultaneously with the composition analyses specified in 4.4.3, a separate 5 g portion from the ground portion of the 1-pound primary sample shall be transferred to a 400 mL beaker. The Potassium nitrate shall be extracted as specified 4.4.3.1, the residue ignited and the ash determined as specified herein.

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4.4.6 Specific gravity.

4.4.6.1 Preferred method (not applicable to classes 7. 8, or 9). A weighed portion of approximately 10 g of the primary sample shall be transferred to a 25 mL specific gravity bottle. Clean Mercury of known temperature, shall be added until the bottle is approximately one half to two thirds full. The remaining air in the bottle shall be displaced with Mercury as follows: A piece of heavy rubber tubing, approximately 2 feet long, shall be attached to the bottle. The other end of the tubing shall be connected to one arm of a Y-tube. By means of another piece of rubber tubing and a short piece of glass tubing, the other arm of the Y-tube shall be extended to dip into a vessel containing Mercury. The stem of the Y-tube shall be connected to a vacuum line. The rubber tubing leading to the Mercury vessel shall be closed by means of a pinchcock. The vacuum line shall be opened in order to evacuate the specific gravity bottle. After evacuation, the vacuum line shall be closed by a pinchcock placed close to the Y-tube. The pinchcock on the tubing leading to the stem of the Mercury vessel shall be opened and the Mercury allowed to flow into the bottle. This operation shall be repeated until the bottle is filled with Mercury. The bottle containing the powder and Mercury shall be weighed. The same bottle, filled with Mercury shall be weighed. The specific gravity shall be calculated as follows:

$$Specific\ gravity = \frac{A\ x\ B}{A + C - D}$$

Where:

A = Weight of sample taken for test
B = Specific gravity of Mercury at test temperature
C = Weight of vessel filled with Mercury
D = Weight of vessel, powder, and Mercury

4.4.6.2 Alternate method (not applicable to classes 7. 8. or 9). In case of dispute or doubt, the preferred method shall be the standard. Add exactly 100 g of primary sample to a clean, standardized bulb. Place the bulb in a vertical position, with the lower stem immersed in the Mercury reservoir, attach the vacuum pump to the upper stem, and proceed to exhaust the air and to fill the space within the bulb with Mercury. Dislodge air bubbles that may cling around the powder grains or to the wall of the bulb by tapping the bulb gently with a rubber stopper while the air is being exhausted. Alternately exhaust the air and allow the Mercury to follow up into the bulb in four repeated operations as described. After finally drawing the Mercury up into the upper stem of the bulb, and after disconnecting the vacuum pump, allow the Mercury to settle back into the bulb. Close both stopcocks, empty both upper and lower stems of all excess Mercury, wipe the exterior globules of Mercury and weigh the bulb containing Mercury and powder. This weight represents the value of "Y" in the formula shown below. Determine the temperature of the Mercury in degrees Centigrade at the time of making the determination. Table VI shows the specified 0 degrees gravity of Mercury "D", at various temperatures ranging from 0 degrees C to 40 degrees C:

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TABLE VI. Specific gravity of mercury.

Temperature											
Degrees	C	0	1	2	3	4	5	6	7	8	9
0	13.5	9545	9298	9051	8804	8558	8311	8065	7818	7572	7326
10		7079	6833	6587	6341	6095	5849	5604	5358	5113	4867
20		4622	4376	3886	3641	3396	3151	2906	2906	2661	2416
30		2171	1927	1682	1437	1193	0949	0704	0460	0216	9972
40	13.4	9728	9484	9240	8996	8752	8508	8265	8021	7778	7534

4.4.6.3 Specific gravity (applicable to classes 7. 8, and 9). Use an apparatus consisting of a Fleming drying jar (see Figure 1) having a capacity of approximately 50 mL. A fritted glass disk of medium porosity is cemented to the bottom of the upper stopcock. The disk should be of the same diameter as the base of the stopcock, beveled so as to have a greater taper than the stopcock, and cemented to the stopcock by means of Canada balsam or equivalent. To fill the jar, attach a piece of rubber tubing to the lower arm of the Fleming jar and immerse the other end in a beaker containing Mercury at room temperature. By means of a piece of heavy rubber tubing connect the upper arm of the jar to a vacuum line and evacuate. Close the upper stopcock, remove the suction tubing, and then close the lower stopcock. Disconnect the tubing from the lower arm, and remove the Mercury from this by tilting and tap the apparatus. Open the lower stopcock so that the Mercury can flow into the side arm if there is any expansion of the Mercury. Weigh the jar and Mercury. Attach to the lower arm a short piece of rubber tubing closed by means of a pinchcock, and allow approximately half of the Mercury to drain from the jar. Using a small funnel, transfer a weighed portion of approximately 10 g of the sample to the jar. Replace the upper stopcock, attach suction tubing to the upper arm, open the upper stopcock, and remove the pinchcock from the rubber tubing while holding the end of the tubing under Mercury in a beaker. Apply suction so that the pressure in the jar is reduced to between 100 and 120 millimeters (mm) of Mercury and all air replaced by Mercury. Gentle tapping of the jar while the vacuum is being applied facilitates the removal of any air bubbles. When the air is replaced, which ordinarily requires approximately 10 seconds, close the upper stopcock, remove the suction tubing and then close the lower stopcock. Remove the rubber tubing and any Mercury remaining in the lower arm. Open the lower stopcock and weigh the jar and contents. Calculate the specific gravity as follows:

$$Specific\ gravity = \frac{A\ x\ B}{A + C - D}$$

Where:

A = Weight of sample taken for test
B = Density of Mercury at room temperature
C = Weight of jar filled with Mercury
D = Weight of jar, powder, and Mercury

Note: See Table VI.

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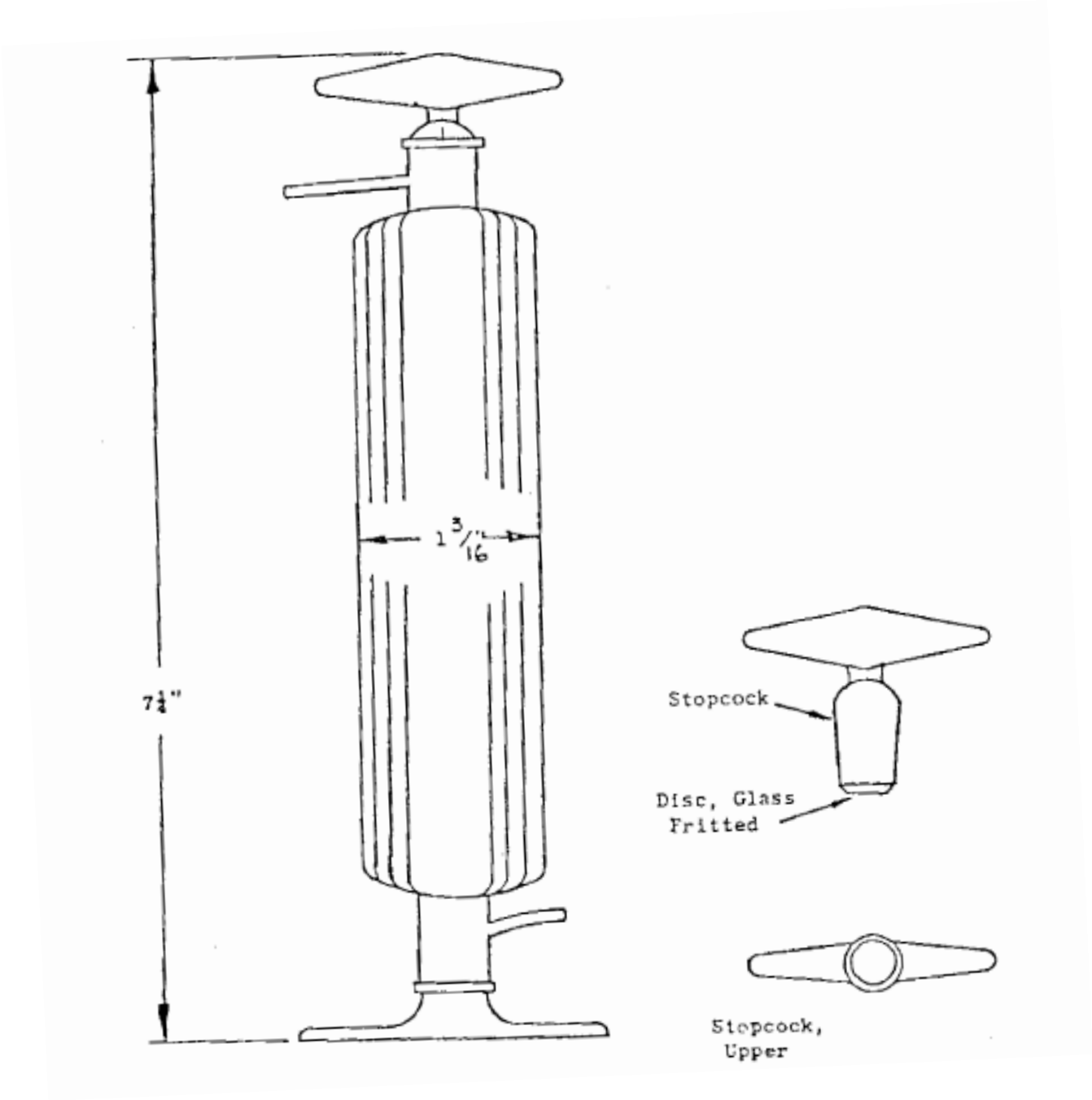


FIGURE 1. Fleming jar.

4.4.6.4 Specific gravity by helium gas pycnometer, (applicable to classes 1 to 7). A helium gas pycnometer (Micromeritics, Accupyc 1330, or equivalent) shall be used. An analytical balance accurate to ± 0.01 mg (Fairfax or equivalent) shall be used. The pycnometer shall be calibrated and operated at the conditions and parameters specified herein. Pycnometer manufacturer instructions and procedures should be followed for the detailed step-by-step operation of the gas pycnometer.

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4.4.6.4.1 Calibration. Calibration shall be performed using factory supplied steel balls (or steel cylinders) with certified values. For refined calibration or trouble-shooting, other certified standards (such as glass bead standards certified by a nationally recognized institute for standards) may be used. The pycnometer shall be calibrated using parameters and conditions specified below:

Helium flow pressure: 20-22 psig (recommended)
Number of purges: 5
Number of runs: 5
Purge pressure: 19.5 to 19.6 psig
Run pressure: 19.5 to 19.6 psig
Equilibration rate: 0.005 psig/min
Use run precision: No. (See note.)
Equipment warm-up: 15 minutes minimum from stand-by and 1 hour minimum from cold start-up.

Note: Run precision may be used for refined calibration or trouble-shooting.

4.4.6.4.2 Procedure for analysis. Sample of black powder shall be acclimated to room temperature before analysis. During winter months, let the unopened sample container warm up to room temperature for at least 16 hours prior to opening the sample container. The black powder sample shall be tested as received. Any action/operation that may cause size attrition or modification, such as grinding or sieving, shall be avoided. The pycnometer shall be checked to verify that it is operating properly prior to sample analysis. An empty sample cup is weighed to the nearest 0.1 mg. Transfer a quantity of sample (about 10 grams) to the cup. The powder must be packed and the cup at least 2/3 full to obtain optimum sample weight for high accuracy analysis. The cup and sample shall be weighed to the nearest 0.1 mg. Place the cup (sample) in the pycnometer chamber and lock the chamber. The sample shall be analyzed following step-by-step operating instructions from the pycnometer manufacturer and at the operating parameters and conditions specified below:

Helium flow pressure: 20 to 22 psig (recommended)
Number of purges: 5
Number of runs: 5
Purge pressure: 16.2 to 16.3 psig. (See note.)
Run fill pressure: 16.2 to 16.3 psig. (See note.)
Equilibration rate: 0.0250 psig/min
Use run precision: Yes
Percent full scale: 0.1

Note: Purge and run pressure of 19.5 to 19.6 psig can also be used for the determination of specific gravity of class 7 black powder.

Calculate the five volumes from the 5 runs, and average the 5 readings. Calculate the specific gravity of the sample by dividing the individual mass of the sample by its corresponding averaged volume.

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4.4.7 Apparent density (see 3.6 for applicability). A 50 mL graduated cylinder, having an internal diameter of approximately 1 inch, shall be weighed to the nearest 0.1 g. The cylinder shall be filled with a portion of the primary sample as close as possible to the 50 mL mark. The cylinder shall be suspended in an upright position by a length of cord, and allowed to fall 30 times from a height of 1 inch onto a rubber cushion. (A standard laboratory rubber stopper, having a diameter slightly larger than that of the base of the graduated cylinder, has been found satisfactory for use). Care shall be taken to prevent any oscillation or sidewise motion. Upon completion of the 30 drops, sufficient powder to raise the level to the 50 mL mark shall be added. The cylinder and contents shall be weighed to the nearest 0.1 mg. The apparent density shall be calculated as follows:

$$Apparent\ density = \frac{A - B}{50}$$

Where:

A = Weight of cylinder plus black powder
B = Weight of empty cylinder

4.4.8 Granulation (not applicable to class 9). The specified sieves for the applicable granulation (see Table III) shall be properly superimposed and assembled with a bottom pan. The sieves shall comply with ASTM E11. A weighed portion of approximately 100 g of the primary sample shall be placed on the topmost sieve of the assembly. The assembly shall be covered and shaken for 3 minutes by hand or by means of mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. The portions retained or passed by the various sieves, as required, shall be weighed and the results calculated to a percentage basis.

4.4.9 Functioning. One hundred percussion primers (M92E1) shall be loaded in accordance with drawing 8863394. The loaded primer shall be placed in a static test fixture and struck by means of a spring loaded firing pin with sufficient energy to activate the element. The primer (see 6.8) shall then be observed for completed burning of the black powder. All primers shall function. If any primer does not function, the first article sample shall be rejected (see 4.2.1).

4.4.10 pH determination. From each sample selected in accordance with 4.3.2.5, a weighed portion of 6.76 ± 0.05 g shall be placed into a 250 mL beaker. Add 100 mL of neutral distilled (or deionized) water. Cover the beaker and digest on a steam bath for 30 minutes. With stirring, cool to 25 degrees C. Determine the pH at 25 degrees C in accordance with ASTM E70. If 25 degrees C is not used, the pH meter shall be recalibrated for that measured temperature.

4.4.10.1 Alternate method. From each sample selected in accordance with 4.3.2.5, a weighed portion of 1.36 ± 0.01 g shall be placed into a 250 mL beaker. Add 100 mL of neutral distilled (or deionized) water. Cover the beaker and digest on a steam bath for 30 minutes. With stirring, cool to 25 degrees C. Determine the pH at 25 degrees C in accordance with ASTM E70. (Test for pH of aqueous solution with glass electrode). If 25 degrees C is not used, the pH meter shall be recalibrated for that measure temperature.

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4.4.11 Workmanship. The black powder shall be uniform in quality and free from lumps, visible impurities, and foreign matter. Failure to meet the workmanship requirement during inspection or production shall result in batch rejection.

5. PACKAGING

5.1 Packaging.

5.1.1 Level A. When specified, black powder should be packaged in easily ignited high strength flax paper bags of either one ounce capacity or four ounce capacity. Either one 4 ounce bag or four 1-ounce bags should be packaged in a plain, bright finish tin can with a slip-on cap. The can shall be sealed with metal foil backed pressure sensitive tape (see 6.5).

5.2 Packing (see 6.2).

5.2.1 Level A or level B.

5.2.1.1 Bulk powder. Bulk black powder shall be packed in steel drums complying with Drawings 7548321 and 7548322. The drums shall be lined with a bag made and closed as specified in 5.2.1.3. When specified, drums shall be lined with muslin liner complying with the Drawing LD299212.

5.2.1.2 Cans. Black powder packaged as specified in 5.1.1 shall be packed in boxes complying with Specifications 14, 15A, or 16A of Department of Transportation (DOT) Regulations CFR 49 Part 100-199.

5.2.1.3 Liners. Liners shall be made of conductive polyethylene or polyolefin sheet with a nominal thickness of .003 in. (tolerance = plus or minus 20%); surface resistivity shall not exceed 1000 ohms per centimeter, when tested in accordance with ASTM D257.

Electrostatic bleed-off shall not exceed 2.00 seconds when tested in accordance with Method 4046 of MIL-STD-3010C. After filling, the top of the bag shall be twisted and the twisted portion folded back on itself; it shall be fastened with a suitable tape or tie. The bag size shall be large enough to prevent strain on the bag if the container is dropped or rolled.

5.2.2 Level C.

5.2.2.1 Metal container. Bulk black powder shall be packed in metal containers complying with Drawings 7548077 or 7548078. The containers shall be lined with a bag made and closed as specified in 5.2.1.3.

5.2.2.2 Alternative metal container. Bulk black powder may be packed in metal kegs of 20 or 25 pounds capacity complying with DOT Specification 13. Kegs shall be lined with a bag made and closed as specified in 5.2.1.3.

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5.2.2.3 Immediate use containers. Black powder intended for immediate use (no more than 180 days storage) may be packaged, packed and marked by any other method approved by DOT Regulations when specifically authorized by the contracting officer. The date of pack shall be clearly marked on the shipping container. The completed pack shall provide sufficient moisture protection to insure that the moisture limit stated in 3.3 will not be exceeded if the black powder is stored up to 180 days under normal storage conditions.

5.3 Marking.

5.3.1 Packaging. The cans specified in 5.1.1 shall have the contents identified on a paper label attached to the body of the can. Additional information shall be as specified by the contracting officer.

5.3.2 Packing. Marking shall be in accordance with applicable drawings. Containers constructed in accordance with Drawing 12982865.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Black powder is intended for the following principal uses:

Class 1	-	JATO and rocket igniters, artillery primers and igniter pads
Class 2	-	JATO and rocket igniters, ignition ends for bag charges, primers, propellant charges for line throwing guns
Class 3	-	JATO and rocket igniters, ejection charges for base ejection shell, and in pyrotechnic items
Class 4 and 5	-	JATO rocket igniters, relay pellets, igniting charges for illumination charges, charges in target practice shell, ejection charges for base ejection shell, igniter charges in primer detonators, and fuze delay elements and tracer igniters
Class 6 and 7	-	JATO and rocket igniters, relay pellets, delay and igniter charges in primer detonators; delay elements, practice hand grenade fuzes, and in Navy squibs
Class 8	-	Propellant charges for rocket signals
Class 9	-	Depth charge cartridges

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Issue of ASSIST to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2)
c. Provisions for submission/resubmission of first article samples.
d. Requirements for contractor Acceptance Inspection Equipment (AIE) design approval (see 6.4).
e. Acceptance and description sheets - Acceptance and description sheets should be prepared for each lot in accordance with MIL-STD-1171.
f. Levels of preservation and packing (see 5.1 and 5.2).
g. Name and address of manufacturer.

6.3 Consideration of data requirements. When this specification is used in a procurement contract which incorporates DD Form 1423, contract data requirements list, acceptance and description sheets (for explosives) are required for each lot. These sheets should be prepared and submitted by the contractor in accordance with MIL-STD-1171 or otherwise as directed by the contract.

6.4 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: CCDC AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.5 Approval of equivalent test method. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to the technical agency (see 6.4). This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision, and drawings of any special equipment required.

6.6 Submission of alternative conformance inspection provisions. Unless otherwise specified in the contract, proposed alternative conformance inspection provisions should be submitted by the contractor for evaluation by the Technical Agency responsible for preparation of this specification (see 6.4).

6.7 Submission of first article samples. Instructions as to the location for evaluation of the first article should be obtained from the contracting officer. Upon receipt of such request, the contracting officer should advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be to: U.S. Army Combat Capabilities Development Command Armaments Center (DEVCOM AC), ATTN: FCDD-ACE-QEE, Picatinny, NJ 07806-5000.

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6.8 Submission of acceptance inspection equipment designs and alternative verification methods for approval. Inspection procedures and equipment designs and proposed alternative verification methods, as required, should be submitted to U.S. Army Combat Capabilities Development Command Armaments Center (DEVCOM AC), ATTN: FCDD-ACE-QEE, Picatinny, NJ 07806-5000. This address should be specified on the Contractor Data Requirements List, DD Form 1423 in the contract.

6.9 Drawings. Drawings prepared prior to the issuance of this Military Specification which specified black powder, may have specified them to be in accordance with Specification A-A-59138. Figure 1 illustrates the correlation between the Army and Navy designations in Specification A-A-59138 and this Military Specification. In addition, the following substitutes should apply for Navy applications:

Drawing or specification requirement	New classes to be used
Cannon	Class 2
Musket	Class 4
FFG	Class 4
Shell	Class 6
FFFG	Class 6
Fuze	Class 7
FFFFG	Class 7
Meal	Class 8

6.10 Intended packaging use.

a. Cans, paragraph 5.1.1. Intended for U.S. Coast Guard use.
b. Muslin liners, paragraph 5.2.1.1. Intended for U.S. Navy use.

6.11 Classification. In accordance with the designation of classifications, paragraph 5-222 of Defense Standardization Manual 4120.3-M, the following designation changes are part of this specification.

"Grade I, II, III and IV" as specified in MIL-DTL-155 is changed to "Grade A, B, C, and D", and Class a, b, c, and d" as specified in A-A-59138 is changed to "Class 1, 2, 3, and 4." In the next revision of the affected specifications, these changes will be implemented.

6.12 Functioning test. Complete procedure for this test may be obtained from the U.S. Army Combat Capabilities Development Command Armaments Center (DEVCOM AC), ATTN: FCDD-ACE-QEE, Picatinny, NJ 07806-5000.

6.13 Incorporated changes. ECP Y22Q2001 provides documentation for noted changes.

6.14 Higher moisture content class 3. Class 3 black powder with a moisture content not exceeding 0.70 percent is available and can be specified in the contract or purchase order.

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6.15 Precautionary note. This specification covers the sampling and testing of toxic and hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures should be followed in the handling and processing of black powder.

6.16 Subject term (key word) listing.

Incendiary composition
Igniter composition
Pyrotechnic
Explosive

6.17 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Army- AR
Navy - OS
Air Force - 70

Preparing Activity:
Army – AR
(Project 1376-2020-004)

Reviewing Activities:
Army –AV, EA, MI, TE
Navy – AS, SH

Civil Agency:
FAS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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